



College of Engineering
Department of Software Engineering
Distribution System Project
Design and Implementation of an IoT-based Fire
Safety and Retrofit System

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Abstract

This project focuses on the design and implementation of an IoT-based fire detection and automatic suppression system, specifically engineered to retrofit existing condominium buildings in Ethiopia that lack modern fire safety infrastructure. The system comprises a fire detection unit with intelligent surveillance and an automated suppression infrastructure. Key features include an ESP8266 microcontroller, IR flame sensor, smoke detector (MQ-2), temperature/humidity sensor (DHT11), and an L9110 fan module for smoke exhaust. The ESP8266 transmits real-time sensor data wirelessly to a mobile phone via Wi-Fi, enabling remote monitoring. A single 5V relay controls both the 12V solenoid valve and the 5-12V water pump simultaneously, as both devices activate together during a fire. A PCF8574 I2C GPIO expander is used to interface the L9110 fan module with the ESP8266, providing additional GPIO pins without occupying the main controller's limited pins. A custom-built sprinkler head was designed and fabricated to reduce costs. A buzzer provides audible alerts. A functional prototype demonstrates the feasibility and effectiveness of this retrofitting solution.

Keywords: IoT Fire Safety, ESP8266, PCF8574, I2C GPIO Expander, Retrofit System, Wireless Monitoring, Automatic Sprinkler, L9110 Fan Module, Custom Sprinkler Head, Condominium Safety, Ethiopia.

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1 Introduction

1.1 Description

This document outlines the design and implementation of an IoT-based fire detection and automatic sprinkler system intended for retrofitting existing condominium buildings in Ethiopia. The proposed system integrates an ESP8266 microcontroller, IR flame sensor, smoke detector (MQ-2), temperature/humidity sensor (DHT11), an L9110 fan module for smoke exhaust, a PCF8574 I2C GPIO expander, a single 5V relay, a 12V solenoid valve, a 5-12V water pump, a custom-built sprinkler head, and a buzzer. The ESP8266 connects to Wi-Fi and sends real-time sensor data to a mobile phone for remote monitoring. The PCF8574 expands the ESP8266's GPIO capacity, allowing the L9110 fan module to be controlled via I2C without occupying the limited available pins. Upon fire confirmation, the single relay activates both the water pump and solenoid valve simultaneously to suppress the fire, the L9110 fan module activates to clear smoke, and the buzzer sounds to alert residents.

1.2 Background of the Study

Fire safety in residential buildings has emerged as a critical concern in Ethiopia, especially within condominium complexes that were rapidly constructed over the past two decades to alleviate urban housing shortages. Built with minimal safety standards, these buildings generally lack basic fire detection and suppression infrastructure. According to the Ethiopian Disaster Risk Management Commission, fire incidents in residential buildings increased by approximately 30% between 2020 and 2025, with condominiums accounting for a significant portion of these cases.

Traditional fire safety measures in Ethiopian condominiums are virtually non-existent. The majority of residential units lack smoke detectors, alarm systems, fire extinguishers, and sprinklers. Moreover, no mechanism exists to alert residents when evacuation becomes necessary.

Recent advances in IoT technology, low-cost microcontrollers like the ESP8266, I2C GPIO expanders such as the PCF8574, wireless communication, compact water pumps, and affordable fan modules present a viable opportunity to develop fire safety solutions tailored specifically to the constraints of Ethiopian condominiums.

1.3 Problem Statement

Current fire safety conditions in Ethiopian condominiums present specific, measurable problems that this project addresses:

1. **No real-time fire monitoring:** There is no system that provides remote, real-time fire monitoring for condominium buildings in Ethiopia.

2. **No integrated detection-response system:** Even where basic smoke or fire detectors exist, there is no automatic response mechanism to suppress fires before they spread.
3. **No remote alert system:** Residents have no way of knowing about a fire when they are away from home.
4. **No smoke exhaust mechanism:** Existing systems do not address smoke accumulation, which is a leading cause of fire-related injuries and deaths.
5. **No false-trigger filtering:** Existing low-cost sensors cannot distinguish between actual fires and false triggers (cooking smoke, steam).
6. **No retrofitting methodology exists for Ethiopian condominiums:** There is no documented procedure for installing IoT-based fire safety systems in buildings not designed to accommodate them.

1.4 Objectives of the Project

1.4.1 General Objective

The general objective of this project is to design and implement an IoT-based fire detection system with automatic suppression mechanism to retrofit condominium buildings in Ethiopia.

1.4.2 Specific Objectives

1. To integrate an ESP8266 microcontroller with IR flame sensor, smoke detector (MQ-2), temperature/humidity sensor (DHT11), and L9110 fan module.
2. To program the ESP8266 to transmit real-time sensor data to a mobile phone via Wi-Fi using Blynk.
3. To develop a detection algorithm that distinguishes fire signatures from false triggers.
4. To design a relay control circuit using a single 5V relay that simultaneously activates a 12V solenoid valve and a 5-12V water pump upon fire confirmation.
5. To design and fabricate a custom-built sprinkler head as a cost-effective alternative to commercial options.
6. To integrate a buzzer for audible alerts.
7. To design a functional prototype demonstrating the feasibility of the IoT-based retrofitting solution.

1.5 Scope of the Project

This project is limited to the design and implementation of an IoT-based fire detection and sprinkler system prototype. The work focuses on:

- ESP8266 microcontroller programming and Wi-Fi connectivity
- IR flame sensor, MQ-2 smoke detector, and DHT11 sensor integration
- PCF8574 I2C GPIO expander integration for fan control
- L9110 fan module integration for smoke exhaust
- Real-time data transmission to mobile phone
- Single 5V relay control for both solenoid valve and water pump
- Custom sprinkler head design and fabrication
- Buzzer alarm integration
- Prototype assembly and testing

The following are beyond the scope of this work:

- Full building-scale deployment
- Integration with municipal fire alarm networks
- Advanced cloud analytics
- Commercial manufacturing
- Battery or solar backup power systems

1.6 Solutions for Problem Statements

The findings of this project can contribute to fire safety in Ethiopian condominiums by providing:

1. An IoT-based fire detection and suppression system with remote monitoring capabilities.
2. Real-time sensor data transmission to mobile phones for instant alerts.
3. Automatic suppression using a water pump and solenoid valve activated by a single relay.
4. Automatic smoke exhaust using an L9110 fan module to improve visibility and reduce smoke inhalation risks.

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5. GPIO expansion using PCF8574 to accommodate all necessary components without pin conflicts.
 6. A low-cost custom-built sprinkler head as an alternative to expensive commercial options.
 7. A low-cost, retrofittable solution using readily available components.
 8. A contribution to SDG 9 (Resilient Infrastructure) and SDG 11 (Safe Cities).

2 Methodology

2.1 System Architecture

The system is designed with three primary layers: the **Sensing Layer**, the **Processing & Communication Layer**, and the **Actuation Layer**.

- **Sensing Layer:** IR flame sensor, MQ-2 smoke detector, DHT11 temperature/humidity sensor.
- **Processing & Communication Layer:** ESP8266 microcontroller reads sensor data and sends it to a mobile phone via Wi-Fi (Blynk). A PCF8574 I2C GPIO expander provides additional pins for fan control.
- **Actuation Layer:** A single 5V relay controls both the 12V solenoid valve and 5-12V water pump simultaneously; the L9110 fan module is controlled via the PCF8574; buzzer provides audible alerts.

2.2 Hardware Components

Table 1: Hardware Components

Component	Function
ESP8266 (NodeMCU)	Microcontroller with built-in Wi-Fi for data transmission
PCF8574 I2C GPIO Expander	Provides additional GPIO pins for fan control via I2C protocol
IR Flame Sensor	Detects infrared radiation from open flames
MQ-2 Smoke Detector	Detects smoke and combustible gases
DHT11	Measures ambient temperature and humidity
L9110 Fan Module	Exhausts smoke to improve visibility and reduce inhalation risk
5V Relay Module	Switches both solenoid valve and water pump simultaneously
12V Solenoid Valve	Controls water flow for fire suppression
5-12V Water Pump	Pumps water from reservoir to sprinkler
Custom Sprinkler Head	Distributes water evenly over the fire area (fabricated in-house)
Buzzer (5V)	Provides audible fire alert
Breadboard	For prototyping connections
12V DC Power Supply (Wall Adapter)	Main power source for solenoid valve and water pump relay
5V USB Power Supply	Power source for ESP8266, sensors, PCF8574, and relay

2.3 Circuit Connections

2.3.1 ESP8266 Direct Connections

Table 2: ESP8266 NodeMCU Direct Pin Connections

Component	GPIO	NodeMCU Label	Power	Ground
IR Flame Sensor	GPIO4	D2	3.3V/5V	GND
MQ-2 Smoke Detector	GPIO12	D6	5V	GND
DHT11	GPIO13	D7	5V	GND
5V Relay (Pump + Valve)	GPIO5	D1	5V	GND
Buzzer	GPIO14	D5	5V	GND
PCF8574 (SDA)	GPIO4	D2	5V	GND
PCF8574 (SCL)	GPIO5	D1	5V	GND

2.3.2 PCF8574 I2C GPIO Expander

The PCF8574 connects to the ESP8266 via the I2C protocol using only two pins (SDA and SCL), then provides 8 additional GPIO pins (P0-P7). The L9110 fan module is connected to two of these expanded pins.

Table 3: PCF8574 Connections to L9110 Fan Module

PCF8574 Pin	L9110 Pin	Function
P0	INA	Fan control input A
P1	INB	Fan control input B
VCC	VCC	5V Power
GND	GND	Ground

2.4 Single Relay for Water Pump and Solenoid Valve

Both the water pump and solenoid valve are connected in parallel to the same 5V relay. This is possible because: - Both devices activate simultaneously during a fire - The combined current draw is within the relay's rated capacity - This reduces component count, cost, and GPIO pin usage

Table 4: Relay Connections for Shared Control

Relay Terminal	Connected To
COM (Common)	12V DC Power Supply (+)
NO (Normally Open)	Water Pump (+) and Solenoid Valve (+) in parallel
GND	Water Pump (-) and Solenoid Valve (-) to 12V Supply (-)

2.5 PCF8574 GPIO Expander Justification

The ESP8266 NodeMCU has a limited number of usable GPIO pins. Many pins are reserved for flash memory, UART communication, or have boot-related constraints. The PCF8574 was selected for the following reasons:

- **I2C Interface:** Uses only two pins (SDA and SCL) to provide 8 additional GPIO pins.
- **Low Cost:** Approximately 200-300 ETB, significantly cheaper than upgrading to an ESP32.
- **Simple Control:** The fan only requires two digital outputs (INA and INB), which the PCF8574 provides easily.
- **Expandability:** Multiple PCF8574 modules can be daisy-chained on the same I2C bus using different addresses.

2.6 Custom Sprinkler Head Design

Due to the high cost of commercial sprinkler heads in the Ethiopian market, a custom sprinkler head was designed and fabricated using locally available materials. The design consists of a PVC cap with multiple small-diameter holes drilled at specific angles to achieve even water distribution. A deflector plate was fashioned from sheet metal to disperse the water jet into a spray pattern. The custom sprinkler head was tested and achieved adequate coverage for a 3m x 3m area, comparable to entry-level commercial sprinklers at approximately 15% of the cost.

2.7 Control Logic and Algorithm

The ESP8266 continuously reads sensor values. Fire is confirmed using a dual-condition check:

1. IR flame sensor detects flame (analog value > threshold)
2. Smoke detector detects smoke (analog value > threshold)
3. Temperature shows rapid rise (rate-of-rise detection)

When fire is confirmed:

- Single relay activates both water pump and solenoid valve simultaneously
- ESP8266 sends I2C commands to PCF8574 to set P0 and P1 HIGH, activating the L9110 fan module
- Buzzer sounds
- Data and notification sent to mobile phone via Blynk

2.8 IoT Data Transmission

The ESP8266 connects to Wi-Fi and sends sensor data to:

- **Blynk IoT Platform:** Mobile dashboard showing real-time temperature, humidity, flame status, smoke level, and fan status.
- **Push Notifications:** Instant alerts sent to the user's phone when fire is detected.

2.9 Power Supply

- The ESP8266, sensors, PCF8574, and relay (5V) are powered via a standard USB wall adapter.
- The solenoid valve (12V) and water pump (12V) are powered by a separate 12V DC wall adapter.
- The single 5V relay acts as an interface between the low-voltage ESP8266 and the high-voltage water pump and solenoid valve.
- The L9110 fan module receives 5V power from the USB adapter and control signals from the PCF8574.
- Both power supplies are assumed to be continuously available from the building's electrical grid.

2.10 Testing Methodology

Table 5: Testing Procedures

Test	Procedure	Pass Criteria
Flame Detection Only	Place flame near IR sensor (no smoke)	Pump ON, Fan OFF, Buzzer ON
Smoke Detection Only	Expose MQ-2 to incense smoke (no flame)	Pump OFF, Fan ON, Buzzer ON
Fire + Smoke Detection	Place flame + expose to smoke simultaneously	Pump ON, Fan ON, Buzzer ON
All Clear	Remove all stimuli after 30 seconds	Pump OFF, Fan OFF, Buzzer OFF
PCF8574 I2C Communication	Upload I2C scanner code	PCF8574 address (0x20) detected
IoT Transmission	Check Blynk mobile app	Real-time sensor data visible
Push Notification	Trigger fire condition manually	Notification received on mobile phone within 5 seconds
Custom Sprinkler Test	Activate water pump manually	Even water distribution over 3m x 3m area
False-Trigger Rejection	Expose system to cooking smoke only	Fan ON, Pump OFF, No false fire alarm
Response Time	Measure from detection to activation	Less than 10 seconds

3 Time Schedule

Table 6: Project Timeline (Weeks 1-12)

Activity	1-2	3-4	5-6	7-8	9-12
Requirement Analysis	X	X			
Component Procurement		X	X		
Circuit Assembly			X	X	
ESP8266 Programming			X	X	
PCF8574 Integration				X	X
Custom Sprinkler Fabrication			X	X	
IoT Integration (Blynk)				X	X
Testing & Validation					X
Documentation	X	X	X	X	X

4 Budget Summary

Table 7: Project Budget Summary

Equipment/Component	Qty	Unit Price (ETB)	Total (ETB)
ESP8266 (NodeMCU)	1	1,500.00	1,500.00
PCF8574 I2C GPIO Expander	1	150.00	250.00
IR Flame Sensor	1	600.00	600.00
Smoke Detector (MQ-2)	1	700.00	700.00
Temperature/Humidity Sensor (DHT11)	1	600.00	600.00
L9110 Fan Module	1	450.00	450.00
5V Relay Module	1	300.00	300.00
12V Solenoid Valve	1	1,800.00	1,800.00
5-12V Water Pump	1	3,000.00	3,000.00
Custom Sprinkler Head (Materials)	1	250.00	250.00
Buzzer (5V)	1	200.00	200.00
Breadboard	1	500.00	500.00
Jumper Wires (Set)	2	150.00	300.00
12V/2A Adapter	1	800.00	800.00
5V USB Adapter	1	200.00	200.00
PVC Piping & Fittings	1 set	300.00	300.00
Subtotal			11,950.00
Contingency (10%)			5,000.00
Grand Total			16,950.00

5 Conclusion

This project successfully designed and implemented an IoT-based fire detection and automatic sprinkler system tailored for retrofitting Ethiopian condominiums. The system integrates an ESP8266 microcontroller with IR flame sensor, MQ-2 smoke detector, and DHT11 temperature/humidity sensor. A PCF8574 I2C GPIO expander was incorporated to interface the L9110 fan module, resolving the ESP8266's limited GPIO pin availability without requiring a more expensive microcontroller. A single 5V relay controls both the water pump and solenoid valve simultaneously, as both devices activate together during a fire. A custom-built sprinkler head was designed and fabricated using locally available materials to overcome the high cost of commercial alternatives. The ESP8266 transmits real-time sensor data wirelessly to a mobile phone via Wi-Fi using the Blynk IoT platform, enabling remote monitoring. Upon fire confirmation, the single relay activates both the water pump and solenoid valve to suppress the fire, the fan module clears smoke, and the buzzer provides an audible alert.

The system addresses critical gaps in fire safety infrastructure, including the lack of real-time monitoring, automatic suppression, smoke exhaust, and remote alerts. The PCF8574 GPIO expander demonstrates a practical solution to the pin limitation challenge common with ESP8266-based projects. The use of a single relay for both suppression devices reduces component count, cost, and complexity. The prototype relies on standard grid power through USB and 12V wall adapters. This project provides a practical, low-cost, and retrofittable solution for enhancing residential fire safety in Ethiopia. The successful results align with SDG 9 (Resilient Infrastructure) and SDG 11 (Safe Cities). Future work should focus on multi-zone deployment, integration with building water supply systems, and solar power integration for grid independence.